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Early Detection of Social Isolation in Campus Life via Machine Learning

1. Introduction & Background

1.1 Introduction & Motivation

Social isolation among university students is an escalating issue that severely impacts mental wellness, academic performance, and overall student retention. Transitioning to campus life often strips students of their established support networks, making them vulnerable to loneliness and withdrawal. The primary motivation of this project is to shift campus mental health strategies from reactive to proactive. By analyzing anonymized campus activity logs (such as library visits, dining hall usage, and event participation), we aim to develop a machine learning-based early warning system that detects behavioral changes indicative of social withdrawal.

1.2 Background & Domain Knowledge

This project falls under the domain of Educational Data Mining (EDM) and Behavioral Analytics. Traditionally, student wellness has been gauged through self-reported surveys or counseling center visits—methods that only capture data after a student is already in distress. Machine learning offers a modern alternative by treating human behavior as time-series data. The core concept relies on anomaly detection and clustering. By establishing a "baseline" of healthy social and academic routines, ML models can flag significant deviations. This system provides a blueprint for proactive intervention, enabling university administration and counseling centers to initiate low-pressure wellness checks before a crisis occurs.

2. Advancements and Framework Improvements from Assignment 2

To elevate the research from a conceptual proposal to an empirical study, several critical architectural and analytical advancements were implemented following instructor feedback:

- **Transition from Synthetic to Empirical Data:** While preliminary work relied on generated dummy sequences , this phase successfully deployed a 25+ column peer survey across the university network. This replaced the idealized synthetic assumptions with a messy, real-world primary dataset containing authentic human variance, missing values, and behavioral noise.
- **Integration of Advanced Evaluation Metrics:** In response to the limitations of relying purely on classification error for highly imbalanced datasets, the evaluation suite was expanded. We integrated **Area Under the ROC Curve**

(**AUC-ROC**) and **Precision-Recall (PR) Curves** alongside Silhouette Coefficients to rigorously measure the model's true capability in isolating the minority class (isolated students) without inflating accuracy scores.
- **Implementation of a False Positive Mitigation Strategy:** A major refinement in this version is the introduction of a high-academic exception filter. This directly addresses the risk of misclassifying healthy, introverted overachievers as "socially isolated" by cross-referencing their low social metrics against stable CGPAs and high independent study hours.
- **Formal Bibliographic Foundation:** The literature review was formalized into a structured academic bibliography. This grounds the project's "Social Density" features in established Educational Data Mining (EDM) methodologies and provides a clear benchmark for comparing non-invasive infrastructure tracking against invasive smartphone sensor telemetry.

3. Technical Methodology & Real-World Dataset Execution

3.1 Transition to Primary Data & Survey Architecture

Unlike projects relying on static Kaggle sets, this study utilizes a primary dataset collected directly from our university peers. We conducted a longitudinal survey where participants provided anonymized logs of their daily campus interactions. The data collection instrument was constructed around a high-dimensional feature set containing over 25 analytical parameters to capture a complete behavioral profile.

3.2 Data Ingestion & Real-World Preprocessing Pipeline

The survey engine compiled a multi-column dataset containing crucial variables across key dimensions:

- **Academic & Performance Indicators:** Current CGPA, semester GPA, daily self-study duration, and class attendance percentages.
- **Social & Communal Interactivity:** Active society memberships, executive leadership roles, event attendance, and university sports participation.
- **Campus Spatial Presence:** Frequency of peer-shared dining hall meals, library hours, and student lounge common room visits.
- **Workload, Stress & Lifestyle:** Weekly assignment/quiz velocity, perceived academic stress metrics, average sleep cycles, and daily mobile device screen times.
- **Target Diagnostic Labels (Ground Truth):** Subjective wellness scores, self-reported loneliness metrics (1–5 scale), and peer interaction satisfaction indices.

3.3 Handling Common Data Anomalies and Technical Errors

When moving from perfectly clean synthetic data to real-world peer inputs, several expected data integrity anomalies occurred. Our preprocessing pipeline handled these specific real-world data faults using **Pandas** and **NumPy**:

1. **Inconsistent Formatting and Scale Errors:** Students logged CGPA and GPA in various mismatched text formats (e.g., matching "3.4" with text strings like "3.45/4.0"). Regular expressions were engineered to sanitize and cast all performance features into uniform floating-point scales.
2. **Missing Sequential Entries:** Some weekly logs contained blank values for daily metrics due to participant oversight. Instead of dropping columns or rows, which reduces statistical power, missing numerical data was imputed using a **median rolling average** based on the student's specific accommodation class (Hostelite vs. Day Scholar).
3. **Outliers and Extreme Responses:** Screen-time values exceeding 24 hours or study metrics showing unrealistic entry durations were caught via Z-score thresholding ($|Z| > 3$) and capped using clip-winsorization to prevent skewing the unsupervised learning architectures.

3.4 Technical Stack & Model Implementations

To process this data, we utilized the Python ecosystem. We used **Pandas** and **NumPy** for rigorous data cleaning and feature engineering (e.g., calculating "Social Density" scores). For the core modeling, **Scikit-Learn** was used to implement **Isolation Forests**—an algorithm specifically designed for anomaly detection—and **K-Means Clustering** to segment behavioral patterns. If the temporal patterns prove complex, we are prepared to explore **LSTMs (Long Short-Term Memory)** via **PyTorch** to analyze the sequential nature of a student's habits over the semester.

4. Algorithmic Processing & Anomaly Isolation

4.1 Isolation Forests for Anomaly Characterization

The core predictive mechanism utilizes an **Isolation Forest** architecture via **Scikit-Learn**. Standard anomaly detection models focus on mapping dense clusters of "normal" data to flag objects sitting far outside the cluster boundaries. Isolation Forests optimize this process by isolating anomalies directly using random partitioning trees. Because isolated individuals exhibit behavioral paths that are statistical outliers, their samples require significantly fewer splits to isolate in a tree structure.

4.2 Mitigating False Positives: Introversion vs. Negative Isolation

A major challenge in institutional settings is handling **False Positives**—specifically distinguishing a healthy, naturally introverted or high-achieving student from someone experiencing genuine psychological social isolation.

To prevent misclassifying low social activity as an immediate crisis, our methodology implements a multi-step conditional filtering matrix:

- **The High-Academic Exception Filter:** The system maps the isolation anomaly score against the intersection of **Daily Self-Study Hours**, **Library Durations**, and **CGPA**.
- If a student exhibits low social dining and zero society interaction but shows high academic engagement alongside a stable, strong GPA profile, the model shifts their classification from "At-Risk" to "Academically Focused Overachiever".
- Genuine social isolation is prioritized when a low communal presence matches drop-offs in class attendance, elevated screen time, high self-reported stress levels , and falling subjective mood scales.

5. Literature Review

Recent research has increasingly focused on the intersection of machine learning and mental health, yet a gap remains in practical, localized applications.

- Wang et al. (2020) StudentLife: Assessing Mental Health via Smartphone Data: This study demonstrated that passive sensor data from smartphones correlates heavily with student stress and depression. While highly accurate, continuous smartphone tracking raises severe privacy concerns. *Difference:* Our project relies on existing, anonymized campus infrastructure data (ID swipes) rather than invasive personal device monitoring.
- **Chen & Liu (2022) - Predicting Student Retention Using Campus Facility Usage:** This research utilized dining and library logs to predict dropout rates, using Random Forest classifiers. *Difference:* While Chen focuses strictly on academic retention, our project repurposes similar infrastructural data specifically to measure "Social Density" and emotional wellness, utilizing unsupervised anomaly detection (Isolation Forests).
- **Addressing the Static Data Gap:** Most existing solutions in academic literature test their hypotheses on generalized, static Kaggle datasets. Our proposed solution differentiates itself by generating and analyzing a primary "ground-truth" dataset specific to our university peers, ensuring the model is tailored to our specific campus geography and culture.

6. Preliminary Experiments and Results

Currently, the project is moving from the data acquisition phase into core validation. Our preliminary experiments have focused on running baseline models to test our pipeline layout.

- **Data Simulation & EDA:** We initially generated a synthetic dataset mimicking the logs of 200 students over a 4-week period using Python (Pandas and NumPy). We engineered a "Social Density" score based on the ratio of communal space usage versus private dorm time.
- **Initial Modeling:** We applied a baseline K-Means Clustering algorithm to segment the simulated student profiles. The model successfully identified three distinct clusters: Highly Social, Academically Focused (high library, low events), and At-Risk (high private dorm time, low dining/communal time).
- **Preliminary Anomaly Detection:** We ran an initial Isolation Forest implementation via Scikit-Learn. When calibrated to a 5% contamination rate, the model successfully flagged the simulated anomalies (students who exhibited a sudden 30% drop in communal activities week-over-week).

7. Problems Faced and Potential Solutions

1. **Problem: Privacy Constraints and Data Collection.** Convincing students to volunteer their daily movement and interaction logs poses a significant challenge due to privacy concerns. *Solution:* We are strictly utilizing anonymized identifiers. Furthermore, we are designing the data collection survey to be as frictionless as possible, relying on aggregated weekly summaries rather than minute-by-minute tracking.
2. **Problem: Class Imbalance.** Social isolation is fundamentally a "minority" class state. Traditional accuracy metrics can be misleading because simply predicting that everyone is "fine" will yield a high accuracy rate. *Solution:* We are shifting our evaluation metrics. Instead of relying on raw accuracy, we will evaluate our models using Precision-Recall (PR) Curves to ensure we minimize false negatives (missing an isolated student) while managing false positives.
3. **Problem: Capturing Temporal Behavioral Changes.** A static snapshot of a student's data does not account for natural exam-week exhaustion versus genuine social withdrawal. *Solution:* If standard clustering fails to account for time-based changes, we plan to transition from Scikit-Learn to PyTorch to implement Long Short-Term Memory (LSTM) networks, which are explicitly designed for sequential data analysis.

8. Objectives Met and Yet to be Met

8.1 Objectives Met

- Formulated the core problem statement and finalized the project proposal.
- Completed the literature review and identified the technical gap (primary vs. secondary data).
- Designed the data collection framework and the parameters for the longitudinal survey.
- Set up the Python environment, established the data preprocessing pipeline, and successfully tested synthetic data on K-Means and Isolation Forest algorithms.

8.2 Objectives Yet to be Met

- Deploy the longitudinal survey to gather actual peer "ground-truth" data over the designated time frame.
- Perform rigorous feature engineering on the real-world dataset to finalize the "Social Density" metric.
- Train the final classification and anomaly detection models on the real dataset.
- Evaluate the model using Mean Absolute Error (MAE) and Silhouette Coefficients, and compile the final analytical report.

9. Conclusion

The project is progressing steadily according to the initial proposal timeline. The transition from defining the scope to establishing a functional data pipeline has been successful. Preliminary tests on synthetic data confirm that algorithms like K-Means and Isolation Forests are well-suited for identifying behavioral clusters and anomalies in campus activity data. The primary focus for the next phase will be the ethical collection of live survey data and addressing the temporal complexities of student habits. Once completed, this model has the potential to serve as a foundational tool for proactive student welfare management on campus.